JavaScript

Working with Http and XHR objects

Peter Kassenaar
info@kassenaar.com

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JavaScript HTTP, or ‘AJAX’

Communicating with web servers

“AJAX:

Asynchrounous JavaScript And XML”

Ajax in pure JavaScript

- AJAX: as a term, it is old now and mostly replaced by simply something like ‘http-communication’ or ‘fetching data’.
- Pure, old school Ajax calls are using the object `XMLHttpRequest`
  - ‘XHR’-object
  - Inventors: IE 5, 1998 (!)
- Pass a `URL`
- Set `RequestHeader`
- Catch event `onreadystatechange`

```
const xhr = new XMLHttpRequest();

// 3. Configure it: GET-request for the URL / random user API
xhr.open('GET', 'https://randomuser.me/api/', true);

// 4. Set up the callback function
xhr.addEventListener('readystatechange', function() {
    // 5. Check if the request is complete (readyState 4) and successful (status 200)
    if (xhr.readyState === 4 && xhr.status === 200) {
        // 6. Parse the JSON response
        const response = JSON.parse(xhr.responseText);
        const user = response.results[0];

        // 7. Display the user data
        const userDataDiv = document.getElementById('userData');
        userDataDiv.innerHTML = `
                <h2>${user.name.first} ${user.name.last}</h2>
                <p><strong>Email:</strong> ${user.email}</p>
                <p><strong>Location:</strong> ${user.location.city},
                                              ${user.location.country}</p>
                <img src="${user.picture.medium}" alt="User Picture">
            `;
    }
};

// 8. Fire/Send the request.
xhr.send();
```

Sample output

Fetching Random User Data

We're using the api `https://randomuser.me/api/` for this example.

Also we are using 'old school' JavaScript `XMLHttpRequest` object here. This way, you build an XHR-request this in older examples, however

Fetch Random User Data

Ricardo Viana

Email: ricardo.viana@example.com

Location: Mogi Guaçu, Brazil

60_XHR_example.html

Ajax/http and a webserver

- NOTE: This does NOT work on the `file://` system. You *will* need a webserver.
- XHR-calls are always being send to a server
  - GET
  - POST
- You can use a `localhost://`-server,
  - or something like `npx serve` from the command line,
  - the current folder is now the root of a webserver / site.
- Lots of examples on StackOverflow, W3Schools, TutsPlus, etc.

http://net.tutsplus.com/articles/news/how-to-make-ajax-requests-with-raw-javascript/

http://www.w3schools.com/ajax/ajax_examples.asp

Final Script

This is a relatively simple script that will allow you to asynchronously request pages by using a "load(URL, CALLBACK)" function.

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```
// Our simplified "load" function accepts a URL and CALLBACK parameter.
load('test.txt', function(xhr) {
    document.getElementById('container').innerHTML = xhr.responseText;
});

function load(url, callback) {
        var xhr;

        if(typeof XMLHttpRequest !== 'undefined') xhr = new XMLHttpRequest();
        else {
            var versions = ["MSXML2.XmlHttp.5.0",
                            "MSXML2.XmlHttp.4.0",
                            "MSXML2.XmlHttp.3.0",
                            "MSXML2.XmlHttp.2.0",
                            "Microsoft.XmlHttp"]

             for(var i = 0, len = versions.length; i < len; i++) {
                try {
                    xhr = new ActiveXObject(versions[i]);
                    break;
                }
                catch(e){}
             } // end for
        }

        xhr.onreadystatechange = ensureReadiness;

        function ensureReadiness() {
            if(xhr.readyState < 4) {
                return;
            }

            if(xhr.status !== 200) {
                return;
            }

            // all is well
            if(xhr.readyState === 4) {
                callback(xhr);
            }
        }

        xhr.open('GET', url, true);
        xhr.send('');
    }
```

w3schools.com

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AJAX Basic
AJAX HOME
AJAX Intro
AJAX Example

AJAX XMLHttpRequest
XHR Create Object
XHR Request
XHR Response
XHR readyState

AJAX Advanced
AJAX ASP/PHP
AJAX Database
AJAX XML File

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Try it Yourself - Examples

A simple AJAX example
Create a simple XMLHttpRequest, and retrieve data from a TXT file.

Load an XML file with AJAX
Create an XMLHttpRequest to retrieve data from an XML file.

Retrieve header information with AJAX
Retrieve header information of a resource (file).

Retrieve specific header information with AJAX
Retrieve specific header information of a resource (file).

Retrieve the content of an ASP file
How a web page can communicate with a web server while a user type characters in an input field.

Retrieve content from a database
How a web page can fetch information from a database with AJAX.

Retrieve the content of an XML file
Create an XMLHttpRequest to retrieve data from an XML file and display the data in an HTML table.

An AJAX example with a callback function
Create a XMLHttpRequest with a callback function, and retrieve data from a TXT file.

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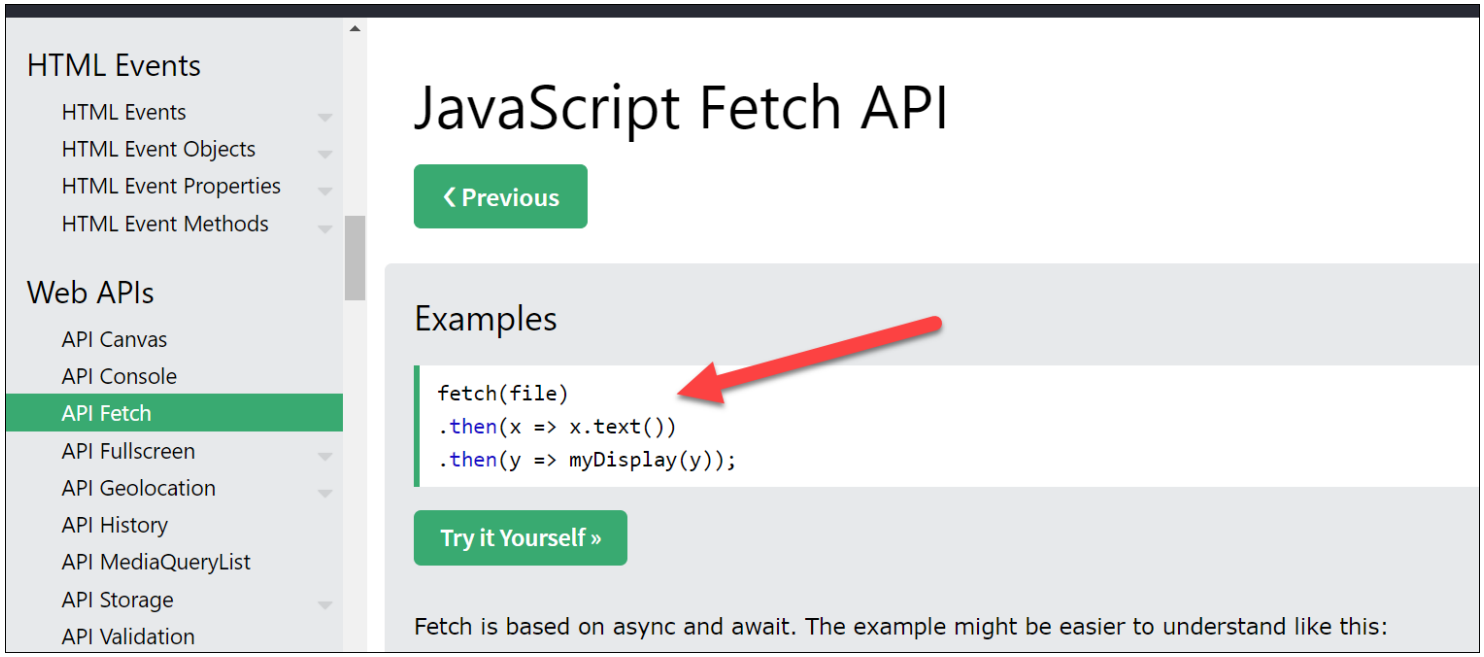
More modern approach – using `fetch()`

- Modern Browsers (Edge, Chrome, Firefox): use `fetch()`.
- Simplifies the process of making HTTP requests in JavaScript.



```javascript
fetch('https://randomuser.me/api/')
    .then(response => {
        if (!response.ok) {
            throw new Error('Error in Network response');
        }
        // Parse the JSON from the response.
        // It is passed to the next .then() in the chain.
        return response.json();
    })
    .then(data => {
        const user = data.results[0]; // only fetch 1 user.

        // Display the user data.
        const userDataDiv = document.getElementById('userData');
        userDataDiv.innerHTML = `
                <h2>${user.name.first} ${user.name.last}</h2>
                <p><strong>Email:</strong> ${user.email}</p>
                <p><strong>Location:</strong> ${user.location.city},
                                              ${user.location.country}</p>
                <img src="${user.picture.medium}" alt="User Picture">
            `;
    })
    .catch(error => {
        console.error('Fetch error:', error);
    });
```

61_fetch.html

Using `async` and `await`

- Instead of using the Promise-notation with `.then()` and `.catch()`, we can also use `async / await` notation
- Functionality is the same!
- `Async / await` is just syntactic sugar over Promises
  - More in line with other languages (C#, Java)
- Some like this notation more.
  - Personal preference

```javascript
// Mark the function as `async`
async function fetchRandomUser() {
    try {
        const response = await fetch('https://randomuser.me/api/');

        if (!response.ok) {
            throw new Error('Error. ');
        }

        // Use the keyword `await` here.
        const data = await response.json();
        const user = data.results[0];

        // 4. Display the user data
        const userDataDiv = document.getElementById('userData');
        userDataDiv.innerHTML = `…`;
    } catch (error) {
        console.error('Fetch error:', error);
    }
}
```

62_async_await.html

Sample output – identical

Fetching Random User Data using `fetch()`and `async/await`

We're again using the api `https://randomuser.me/api/` for this example.

This script is using the more modern `fetch()` again, but this time not in a Promise notation, but with `async` and `await`. Some like this notation more, but there are no functional differences.

Fetch Random User Data

Anand Dhamdhame

Email: anand.dhamdhame@example.com

Location: Ujjain, India

Server sided – PHP, Java, .NET, etc.

- Make sure server returns correct `MIME Type`!
- You need to configure and format the return results yourself!
  - Based on your server sided platform
- For instance: PHP:

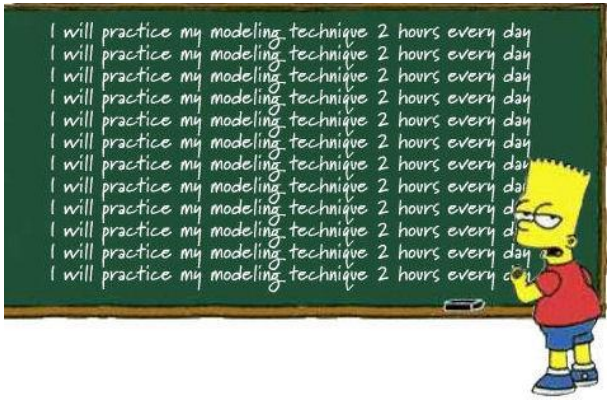
```
<?php
    header('Content-type: application/json')
?>
<?php
    // Zorg hier voor het ophalen & formatteren
    // van gegevens uit de database. Let op
    // de correcte JSON-notatie met aanhalingstekens,
    // dubbele punt en komma's.
?>
{ "naam": "peter", "leeftijd" : 43, "woonplaats": "Dieren" }
```

ASP.NET

- Use for instance `ASP.NET Web API`
- Use `Newtonsoft.json DLL`

Workshop #40

- Create your own JavaScript-http call to the randomuser.me api
  - Or pick an API you know and like
- Load external data into your page
  - Results can be text, HTML or JSON, depending on the API
- Display all data in a loop on screen.



Sample output

JavaScript- fetching Random User Data

Fetch Random Users

1 - Jovana Rodić

Email: jovana.rodic@example.com

Location: Zaječar, Serbia

2 - Elizabeth Anderson

Email: elizabeth.anderson@example.com

Location: Belmont, Canada